

A Beginner's Primer

Understanding the Apollo 15 & 17 Lunar Heat-Flow Study

Companion to “Difference of Lunar Regolith Thermal Conductivity K_d at the Apollo 15 and 17 Heat-Flow Boreholes”

June 15, 2026

How to read this document

This primer assumes *no* background. It builds up from “why does anyone care about heat in moon dirt” to “what our numbers mean and where they could still be wrong.” Coloured boxes flag the two things beginners most want: a **plain-words** restatement, and a **term-by-term** reading of each equation. You can skip the equation boxes on a first pass and still follow everything.

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1 The big picture: what we measured and why

The Moon has no air and no weather. The only thing that moves heat around inside it is *conduction* — heat seeping through solid material, the way a metal spoon left in hot soup slowly warms at the handle. How fast heat seeps through the Moon’s soil (called *regolith*: broken rock and dust) is set by a single number, the **thermal conductivity**. A big conductivity means heat passes easily; a small one means the soil is a good insulator (the Moon’s regolith is one of the best insulators known — better than styrofoam).

Why does this number matter?

- Spacecraft orbiting the Moon measure its surface temperature and use it, plus an assumed conductivity, to infer what the ground is like. If the assumed conductivity is wrong, those inferences are wrong.
- Future missions that “see” below the surface (radar, microwave, terahertz sounders) need to know the temperature *at depth* to interpret their signals. That deep temperature depends directly on the conductivity.

Here is the problem this study tackles: today, scientists use *one single conductivity value for the entire Moon*. Nobody has ever checked that assumption against direct underground measurements — because those measurements exist at only **two places in history**.

In plain words

We use the only two underground temperature records ever collected on the Moon — from the Apollo 15 and 17 missions — to ask: *is one Moon-wide conductivity good enough, or do different places need different values?* Our answer: the two sites genuinely differ, by roughly a factor of two.

2 The physics, gently

2.1 Heat flowing in a column of soil

Picture a vertical column of lunar soil (Figure 1a). Two things heat and cool it:

1. **From above:** sunlight bakes the surface during the two-week lunar day; at night the surface cools by glowing away its heat as infrared light. This makes the surface temperature swing enormously — roughly from -150 to $+100^\circ\text{C}$.
2. **From below:** the Moon’s interior is still warm from its formation and from radioactive decay, so a tiny, steady trickle of heat flows *upward* through the column. We call this the **basal heat flux**, Q_b .

2.2 Why the daily swing disappears with depth

The day–night temperature wave does not reach very far down. Each step deeper, the swing shrinks (Figure 1b). By about a hand’s-width below the surface it is already tiny; by 80 cm it is smaller than we can even measure. The depth over which the swing fades is called the **skin depth**.

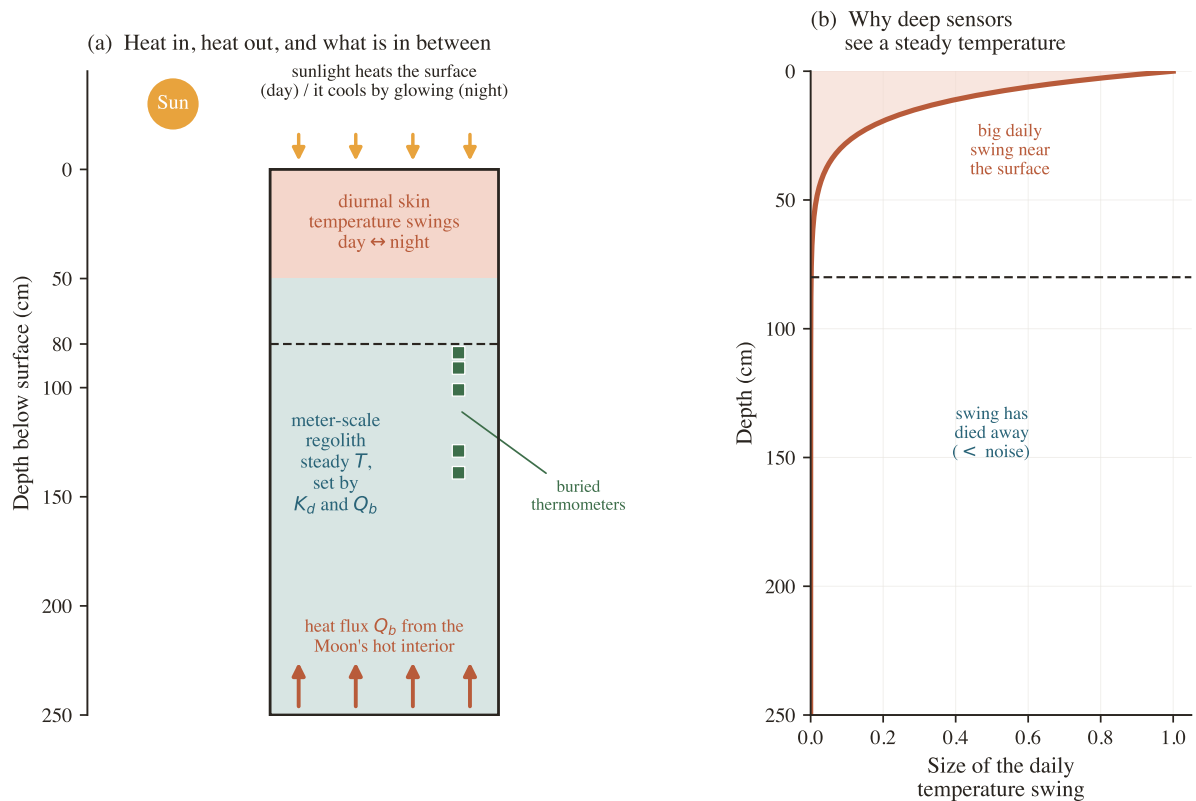


Figure 1. How heat moves in the lunar soil. (a) Sunlight heats the surface from above; a steady flux Q_b rises from the warm interior below. Near the surface (coral band) the temperature swings with day and night; deeper down (teal) it settles to a steady value set by the conductivity K_d and the flux Q_b . The squares mark where the Apollo thermometers actually sat. (b) The size of the daily swing shrinks extremely fast with depth — by ~ 80 cm it is far smaller than the measurement noise, which is exactly why the deep sensors read a steady temperature.

In plain words

Think of pressing a warm hand on a thick blanket. The top fibres warm up, but a few centimetres in, nothing changes. The Moon's daily heat "hand-press" only penetrates the top few centimetres; everything deeper just feels the steady average.

This is the key that makes the whole study possible: **below the skin depth, the soil is in a steady state**. The temperature there does not wobble — it just increases slowly and steadily with depth, because heat is trickling up from below. And the steepness of that increase is controlled by the conductivity.

2.3 The one relationship at the heart of everything

In the steady deep region, the upward heat flux Q_b , the conductivity K_d , and the temperature gradient (how many degrees the temperature rises per metre of depth) are tied together by one simple law:

The equation, term by term

$$\underbrace{\frac{\Delta T}{\Delta z}}_{\text{gradient}} = \frac{\overbrace{Q_b}^{\text{heat from below}}}{\underbrace{K_d}_{\text{conductivity}}}$$

$\Delta T/\Delta z$ — how fast temperature rises with depth (degrees per metre). Q_b — the steady heat flux from the interior (a known, measured quantity). K_d — the deep conductivity, the thing we want to find.

Read it like this: if you push a fixed amount of heat (Q_b) up through a good insulator (small K_d), the temperature has to climb steeply to force that heat through — a big gradient. Through a good conductor (large K_d), the same heat slides through easily and the temperature barely changes — a gentle gradient.

In plain words

We *measure* the gradient (from the buried thermometers) and we *know* the heat flux Q_b . So we can solve for the conductivity K_d . That is the entire idea of the study in one sentence.

2.4 The full equation the computer actually solves

The simple gradient law above is the steady-state shortcut. To handle the day–night swing too, the computer solves the general **heat equation**, which just says “temperature changes in time because heat flows in and out”:

The equation, term by term

$$\underbrace{\rho c_p \frac{\partial T}{\partial t}}_{\text{heat stored here}} = \underbrace{\frac{\partial}{\partial z} \left(K \frac{\partial T}{\partial z} \right)}_{\text{heat flowing in minus out}}$$

T temperature; t time; z depth. ρ density and c_p specific heat: together they say how much heat it takes to warm the soil. K conductivity. The left side is how fast a bit of soil heats up; the right side is the net heat arriving by conduction. In the deep steady state the left side is zero and this reduces back to the gradient law above.

There are two boundary conditions — rules at the two ends of the column. At the **top**, the surface temperature is whatever makes “sunlight in = infrared glow out + heat conducted downward” balance at every instant. At the **bottom**, the heat flux equals Q_b . That bottom rule is important and is one thing this study insists on: the column is fed heat from below, not sealed off.

3 The data: two boreholes, fifty years old

In 1971 and 1972 the Apollo 15 and 17 astronauts drilled holes and lowered strings of thermometers more than two metres down (Figure 2). These “Heat-Flow Experiment” (HFE) probes recorded temperature for years. The records were partly lost, then *restored from the original 1970s tapes* in 2018 — that restored record is what we use.

Two practical points about the data:

- **We use only the deep sensors (below 80 cm).** The fibreglass stem itself conducts a little heat down from the surface, contaminating the shallowest thermometers. Rather than

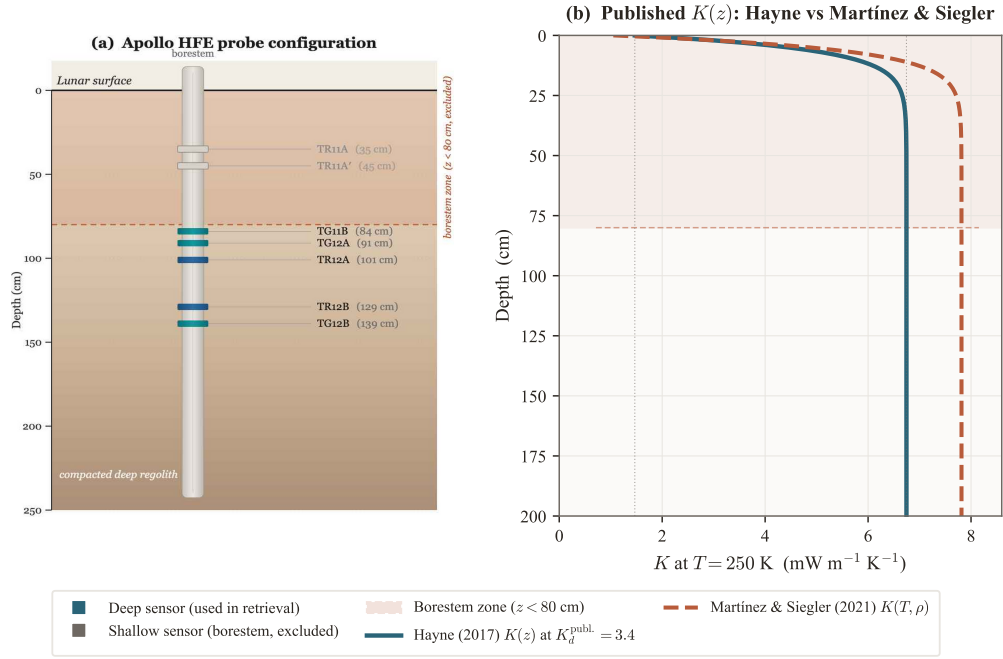


Figure 2. What was buried. The Apollo probe: a fibreglass stem carrying thermometers at several depths. The shaded band near the top ($z < 80$ cm) is the zone we throw away (see below). The right panel shows the two textbook conductivity models we compare against.

try to correct that, we discard everything above 80 cm *before looking at the answer*. This is the “borestem cut.”

- **We use a steady, settled temperature for each sensor.** Each record has early wobbles (from the drilling, equipment hiccups, slow drift). For each thermometer we automatically find the long, flat, settled stretch at the end and average it. That single settled value per sensor is what we compare the model to.

In plain words

The raw material is: for each buried thermometer that sits deeper than 80 cm, one trustworthy “settled” temperature and the depth it sits at. Apollo 15 gives 7 such points; Apollo 17 gives 16. That is the entire dataset — 23 numbers. Everything rests on getting the most out of them honestly.

4 How the code turns temperatures into a conductivity

4.1 The retrieval, step by step

We cannot measure conductivity directly. So we do what scientists do when a quantity is hidden: *guess it, simulate the consequences, and see which guess matches reality* (Figure 4a).

1. Take the real settled temperatures from the sensors (the data).
2. Pick a trial value of K_d . Run the heat-equation simulation of the Moon with that value; it predicts a temperature at every depth.
3. Measure the mismatch between prediction and data with a single number, the **RMSE** (root-mean-square error — essentially the typical disagreement, in degrees).

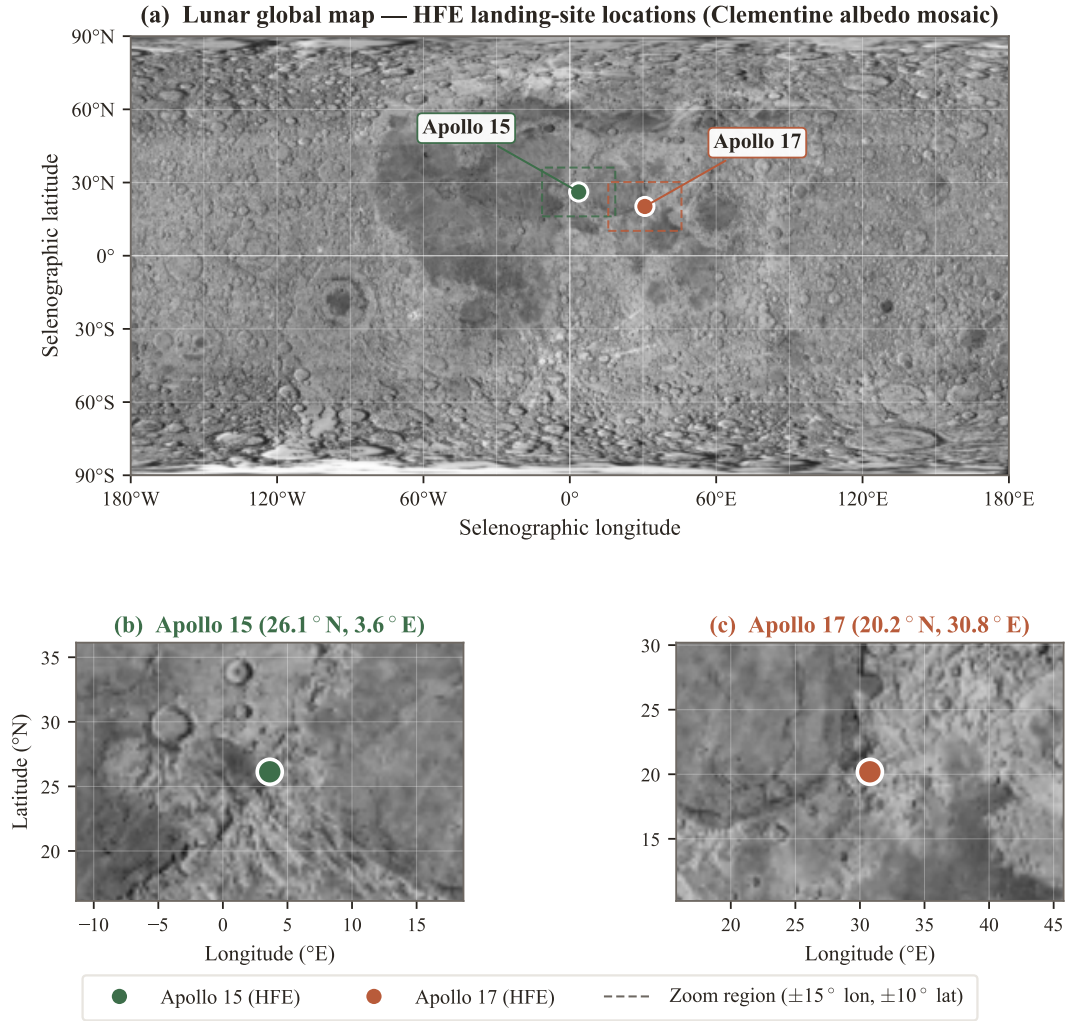


Figure 3. Where the two boreholes are. Apollo 15 sits on a boundary between dark lava plains and bright highlands; Apollo 17 sits in a lava-filled valley. Different geology is one reason the two sites might genuinely have different conductivities.

4. Repeat for many trial K_d values.
5. The trial that gives the smallest mismatch is our answer, written K_d^* .

The equation, term by term

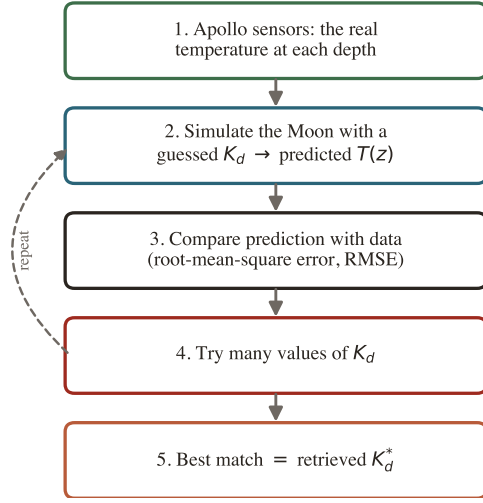
$$\text{RMSE}(K_d) = \sqrt{\frac{1}{N} \sum_{i=1}^N (T_i^{\text{model}} - T_i^{\text{data}})^2}$$

For each of the N sensors, take the gap between model and data ($T_i^{\text{model}} - T_i^{\text{data}}$), square it (so over- and under-shoots both count as error), average over all sensors, and take the square root to get back to degrees. Small RMSE = good fit.

4.2 Putting an error bar on the answer

The bottom of the bowl gives one number, but how sure are we? With only 7 or 16 data points, we use a technique called the **bootstrap**: randomly resample the sensors (with small random

(a) How a conductivity number is extracted



(b) The best fit is the bottom of the bowl

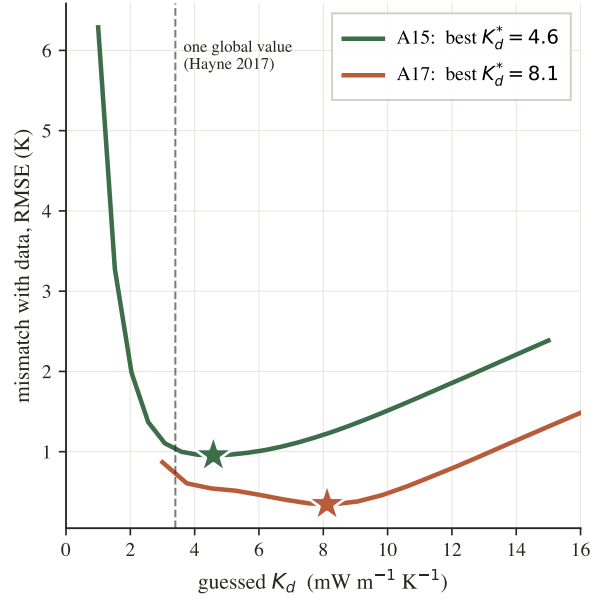


Figure 4. How a conductivity number is extracted. (a) The guess-simulate-compare loop. (b) The real result: plotting the mismatch (RMSE) against the trial K_d traces a bowl, and the bottom of the bowl is the best-fit conductivity. Apollo 15’s bowl bottoms near 4.6, Apollo 17’s near 8.1 — the dashed line is the single “global” value everyone currently uses (3.4).

nudges to their depths, reflecting how precisely the astronauts placed them), redo the retrieval, and repeat 1,500 times. The spread of answers is our uncertainty (Figure 5).

5 The mistake we found and fixed

This is the most important section for understanding why the numbers in the latest version differ from earlier drafts.

5.1 What went wrong

To start a simulation, the computer needs an initial temperature for the deep soil. The old code simply *typed one in* (250 K for Apollo 15, 255 K for Apollo 17) and then let the simulation run for 30 lunar days to “settle.” The trouble: deep lunar soil is such a good insulator that it takes the equivalent of *roughly a thousand* lunar days to truly settle. Thirty is nowhere near enough. So the deep temperature the model reported was really just the typed-in starting guess, barely changed — and the conductivity we then read off was partly a reflection of that guess, not of the Moon.

We proved this directly (Figure 6a): running the old method from three different starting guesses gave three different deep temperature profiles, fanning apart by several degrees. A result that changes when you change an arbitrary starting guess is not yet a measurement.

5.2 How we fixed it

We replaced the “start from a guess and wait” approach with one that computes the true settled state directly, using the physics shortcut from Section 2.3. The trick: in a settled column, the same heat must flow through every layer, and that fact alone fixes the whole deep temperature profile — no waiting required. The method alternates between a short simulation (to settle the

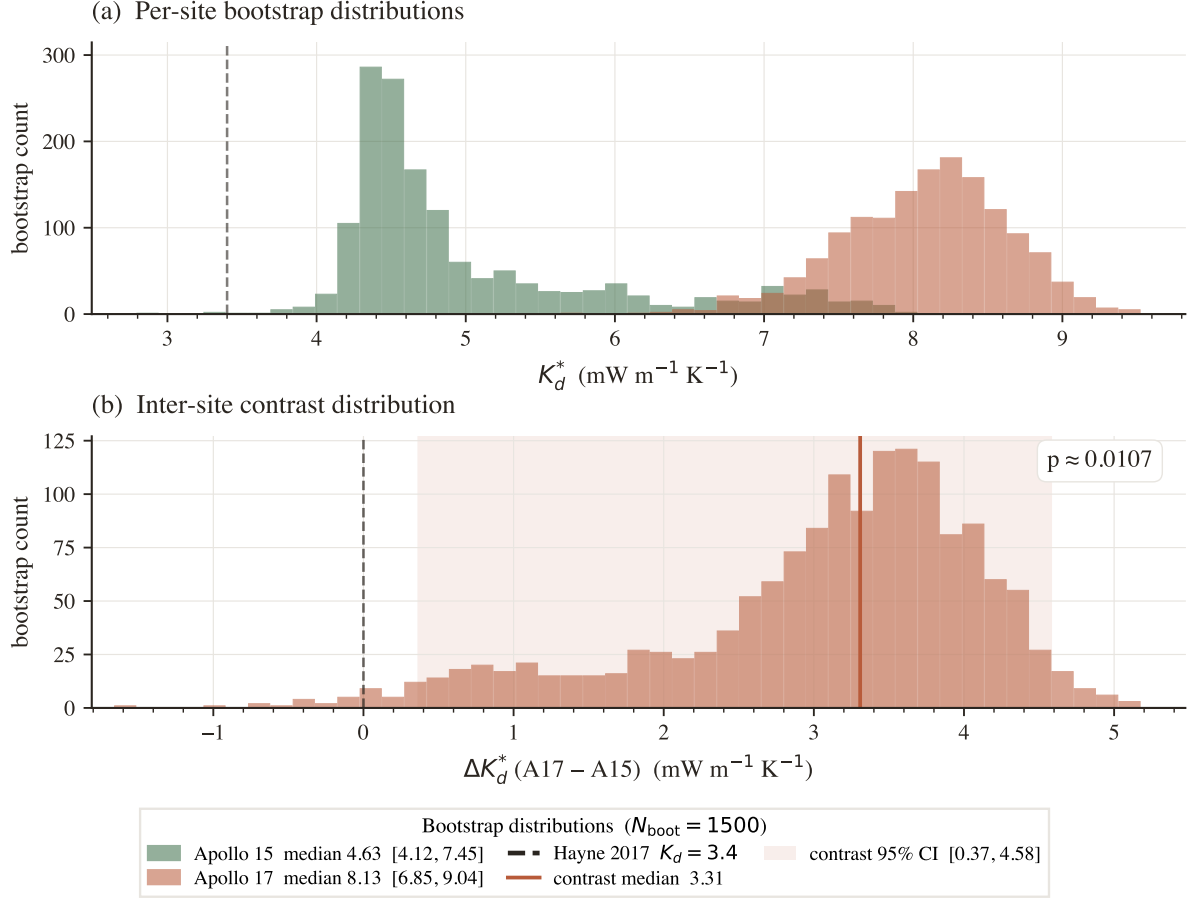


Figure 5. How confident are we? (a) Running the retrieval 1,500 times on resampled data gives a spread of answers for each site — two clearly separated humps. (b) The difference between the sites is almost always positive (Apollo 17 more conductive); it lands on the wrong side of zero only about 1% of the time, which is what “statistically significant” means here.

fast surface skin) and a direct calculation (to fix the slow deep part), repeating until it stops changing.

In plain words

We did *not* simply “run longer.” Running a thousand lunar days for every one of thousands of trial values would take days of computer time. Instead we used the steady-state heat-balance shortcut to jump straight to the settled answer, at the cost of the old short run.

Crucially, we *verified* it works (Figure 6b): starting from wildly different guesses (240 K vs 260 K) now lands on the same answer to within two-hundredths of a degree, and a long check-run started from the converged state does not drift. The result is now a property of the data and the physics, not of a typed-in number.

5.3 What changed in the numbers, and why

- **Apollo 15:** 4.86 → 4.58. Barely moved — its old starting guess (250 K) happened to be close to the truth.
- **Apollo 17:** 11.23 → 8.12. Dropped a lot — its old guess (255 K) was too warm, which had inflated the old conductivity. With the honest deep temperature, the inflation disappears.

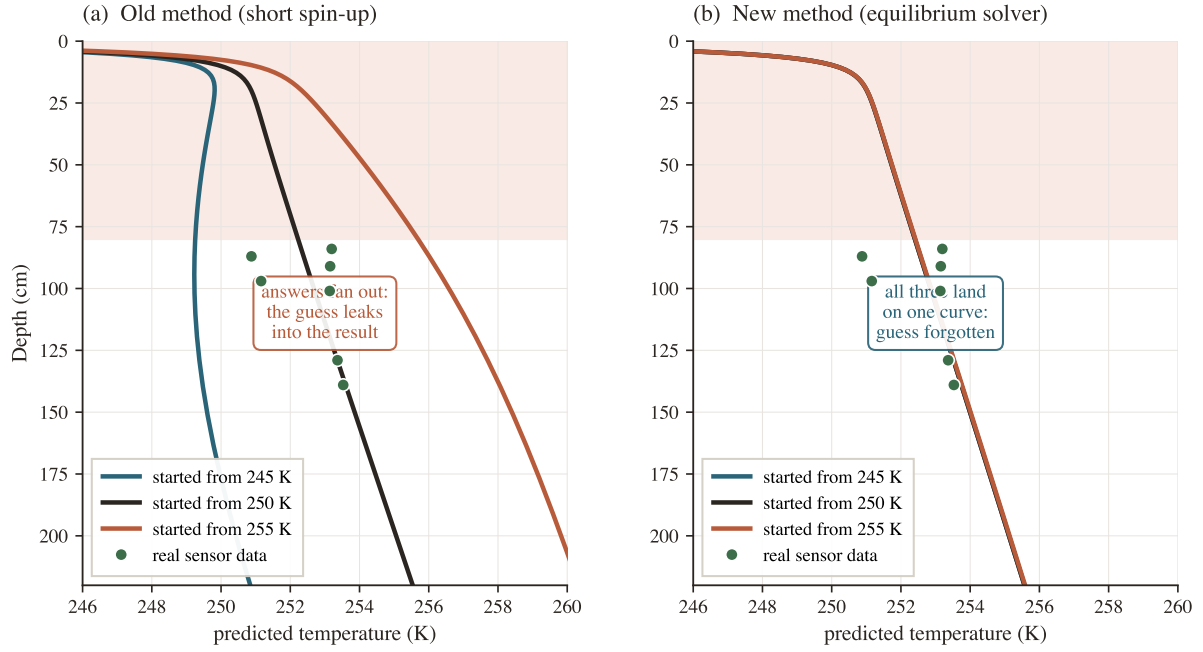


Figure 6. The bug and the fix (real model runs, Apollo 15). (a) Old method: start the simulation from 245, 250, or 255 K and the predicted deep temperature fans out — the arbitrary starting guess leaks into the answer. (b) New method: the same three starts collapse onto a single curve that passes through the real data. The answer no longer depends on the guess.

In plain words

The fix made the science *stronger*, not weaker. The difference between the two sites became statistically significant for the first time, the Apollo 17 value moved into the range that other lines of evidence expect, and an independent conductivity model that used to *contradict* our Apollo 17 result now *agrees* with it. When removing a bug makes several independent checks line up, that is the signature of a correct fix.

6 The results and what they mean

6.1 The headline numbers

	Apollo 15	Apollo 17
Retrieved K_d^* ($\text{mW m}^{-1} \text{K}^{-1}$)	4.58	8.12
95% confidence interval	[4.1, 7.5]	[6.9, 9.0]
Mismatch vs. one global value (RMSE)	1.03 $\rightarrow$ 0.95 K	0.69 $\rightarrow$ 0.34 K

(The unit $\text{mW m}^{-1} \text{K}^{-1}$ is milliwatts per metre per degree — the standard way to quote conductivity. For comparison, water is about 600 in these units and styrofoam about 30; lunar regolith near 5–8 is an extraordinary insulator.)

The two profiles, with the best-fit models drawn through the data, are shown in Figure 7.

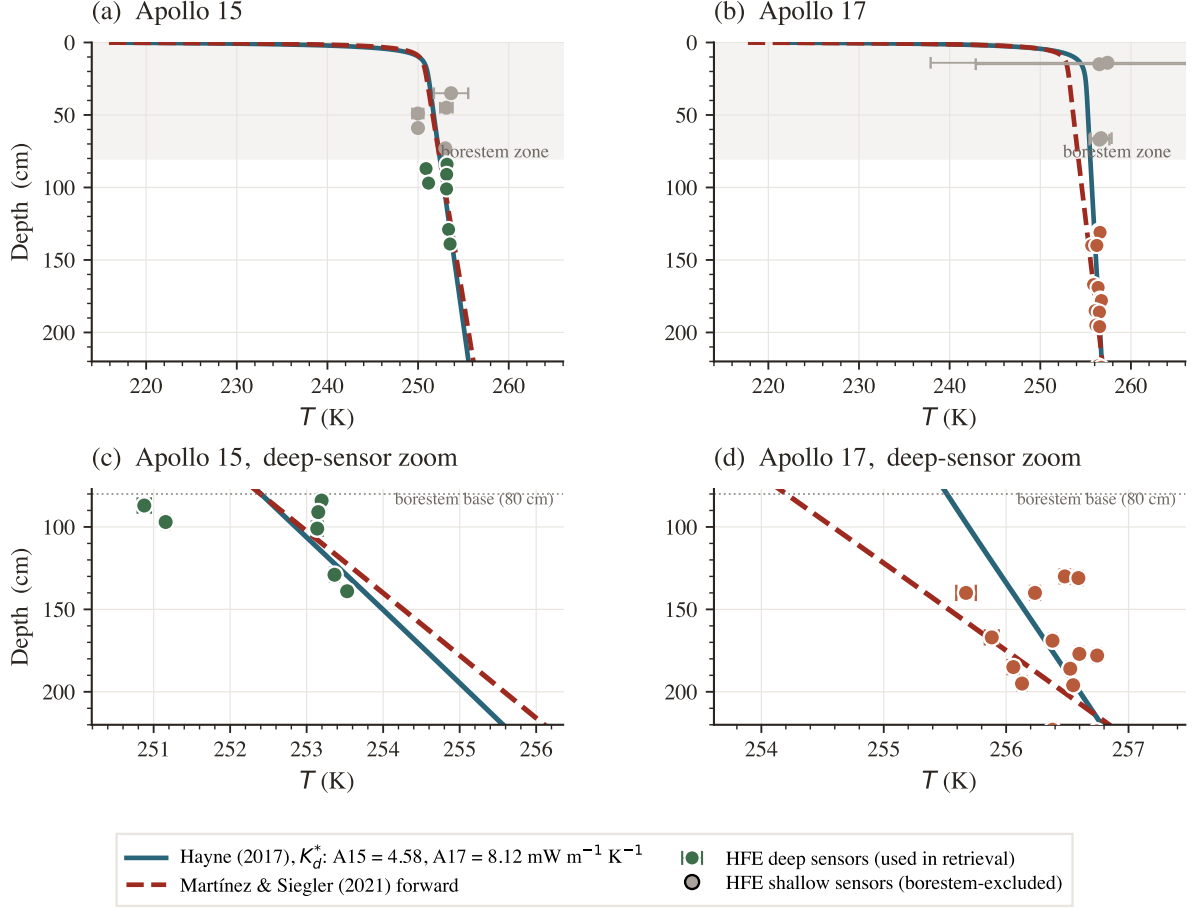


Figure 7. The fitted temperature profiles. Coloured dots are the real deep sensors; curves are the best-fit models. The bottom row zooms in on the deep region actually used. Apollo 15 shows the *steeper* temperature rise with depth — the signature of its lower conductivity (and higher heat flux), since a poorer conductor needs a steeper gradient to pass the heat (the gradient law, Section 2.3). Apollo 17, more conductive, rises more gently.

6.2 Why these particular values make physical sense

A conductivity of 4.6 at Apollo 15 sits comfortably between the global average (3.4) and orbital estimates (~ 7), and matches what soil porosity (how fluffy vs. packed it is) predicts.

Apollo 17 being ~ 1.8 times higher is explained by two effects that both point the same way: (1) its soil is volcanic basalt, which is intrinsically more conductive than the highland rock at Apollo 15; and (2) it appears slightly more tightly packed. Neither effect has to be pushed to an extreme — together they comfortably cover the factor of 1.8.

In plain words

The bigger conductivity at Apollo 17 is not mysterious: it is exactly what you would expect from denser, more volcanic soil. Two independent soil-property arguments predict the size of the difference we measured.

6.3 An independent sanity check

The retrieval only used *deep* temperatures. As a completely separate test, we checked whether the same models reproduce the *surface* temperature swing measured by an orbiting spacecraft

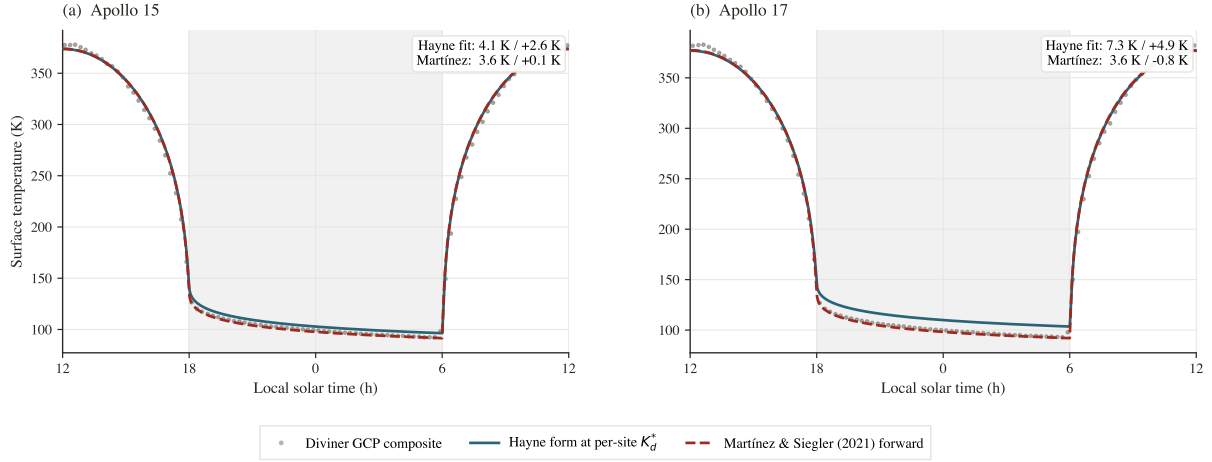


Figure 8. Independent check. Grey dots: surface temperature over a lunar day measured from orbit. Curves: our models. The good match (a few K over a 280 K swing) confirms the forcing is right; this data was not used in the retrieval.

(Diviner). They do (Figure 8) — confirming the whole simulation chain (sunlight, day length, surface balance) is set up correctly, independently of the deep retrieval.

7 Honest limits (what a critic would say)

A good study states clearly where it could be wrong. Ours has three main caveats, all in the manuscript:

1. **The absolute values lean on the assumed heat flux Q_b .** The gradient law fixes the *ratio* Q_b/K_d , not K_d by itself. We used the standard published Q_b ; if those are revised, our K_d values shift proportionally. The *difference* between the sites is far more robust than either absolute value.
2. **Only two boreholes exist.** Two points cannot prove that conductivity varies smoothly across the whole Moon — only that these two specific places differ. Settling the broader question needs new landed missions.
3. **The Apollo 17 value is sensitive to a few inputs.** Its precision (the error bar) is tight, but its accuracy depends on choices like the surface brightness coefficient; we report this openly in the error budget.

In plain words

What is rock-solid: the two sites need different conductivities, and Apollo 17 is the more conductive one. What is conditional: the exact numbers, which depend on the assumed heat flux and the chosen model form. The paper is careful to say which is which.

8 How to run the code yourself

The whole study regenerates from the public repository with a few commands:

```
git clone https://github.com/rp3gregorio/Lunar-HFE.git
cd Lunar-HFE
python3 -m venv .venv && source .venv/bin/activate
```

```
pip install -e "[dev]"
```

```
# the core retrieval (writes output/kd_retrieval_results.json):  
python scripts/pipeline/retrieve_kd.py
```

```
# the figures:  
python scripts/figures/make_letter_figures.py  
python scripts/figures/make_results_figures.py
```

A tour of the pieces:

- `lunar/` — the engine. `solver.py` steps the heat equation in time; `equilibrium.py` is the settled-state solver from Section 5; `properties.py` holds the conductivity/density/heat formulas; `constants.py` stores every physical number with its literature source.
- `scripts/pipeline/` — the analysis: the retrieval, bootstrap, and all the sensitivity checks.
- `scripts/figures/` — the figure generators (including `make_primer_figures.py`, which made the teaching figures in this document).
- `output/` — the committed numerical results, so anyone can verify the paper’s numbers without rerunning.
- `docs/FLAG_REPORT.md` — the full record of the audit and the bug fix described in Section 5.

Glossary

Regolith The Moon’s surface layer of broken rock and dust.

Thermal conductivity (K , K_d)

How easily heat passes through material. K_d is the value deep down, where the soil is packed and steady. Small = good insulator. Our target quantity.

Basal heat flux (Q_b)

The steady trickle of heat rising from the Moon’s warm interior.

Diurnal skin / skin depth

The shallow zone (top few cm) where the day–night temperature swing is still felt; the skin depth is how far it reaches.

Gradient ($\Delta T/\Delta z$)

How many degrees the temperature rises per metre of depth. Steep gradient = good insulator.

Heat equation The differential equation governing how temperature evolves as heat flows; what the solver integrates.

Boundary conditions

The rules at the two ends of the column: radiation balance at the surface, fixed heat flux Q_b at the bottom.

Retrieval The procedure of guessing K_d , simulating, and keeping the value that best matches the data.

RMSE Root-mean-square error: a single number for the typical model–data disagreement, in degrees. Smaller is better.

Bootstrap	Resampling the data many times to measure how uncertain the answer is.
Equilibrium / steady state	The settled condition where the temperature no longer changes from one lunar day to the next.
Borestem cut	Discarding sensors above 80 cm because the probe stem contaminates them.
HFE	Heat-Flow Experiment — the Apollo instrument packages whose data we use.
A15 / A17	Shorthand for the Apollo 15 and Apollo 17 sites.